

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2 – COMPARATIVE ANALYSIS**

Computer Architecture

Combined Assessment Sheet + Answer Document

B.E. ECE

*Prepared as a single combined PDF*



**SIMATS ENGINEERING**  
**Department of Computer Science & Engineering**  
**ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS**

|                |                                                                                                                                                                                                                                                    |               |                                           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------|
| Course Code:   | CSA12                                                                                                                                                                                                                                              | Course Title: | Computer Architecture                     |
| CO Assessed:   | CO4                                                                                                                                                                                                                                                | Assessment:   | ASSESSMENT TOOL2-<br>COMPARATIVE ANALYSIS |
| Weightage:     | 30%                                                                                                                                                                                                                                                | Total Marks:  | 25                                        |
| CO4 Statement: | Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics. |               |                                           |
| Student Name:  | R. DINESH KUMAR                                                                                                                                                                                                                                    | Reg.No.:      | 192312258                                 |
| Department:    | B. E ECE                                                                                                                                                                                                                                           | Date:         | 26/08/2026                                |

**ANNEXURE A**  
**COMPARATIVE ANALYSIS**

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

**Code of Conduct:**

(192312258)

I R. DINESH KUMAR (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

*R. Dinesh Kumar*

**MARKS ALLOCATION**



|    | Scope of Items<br>(20%) | Comparison<br>Depth (25%) | Use of<br>Evidence<br>(20%) | Critical Thinking<br>(20%) | Clarity of<br>Presentation (15%) |   |
|----|-------------------------|---------------------------|-----------------------------|----------------------------|----------------------------------|---|
| Q1 | 0.5                     | 1.25                      | 1                           | 0.5                        | 0.75                             | 4 |
| Q2 | 0.5                     | 1.25                      | 1                           | 0.5                        | 0.75                             | 4 |
| Q3 | 1                       | 0.5                       | 0.5                         | 0.5                        | 0.5                              | 3 |
| Q4 | 1                       | 1                         | 0.5                         | 1                          | 0.5                              | 4 |
| Q5 | 1                       | 1                         | 0.5                         | 1                          | 0.5                              | 4 |

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# RUBRICS FOR CALCULATION:

| Criteria                | Weightage | Level 4 – Exemplary                                                           | Level 3 – Proficient                            | Level 2 – Developing                       | Level 1 – Beginning                      |
|-------------------------|-----------|-------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------|------------------------------------------|
| Scope of Items          | 20%       | Selects highly relevant items with clear scope and justification              | Selects appropriate items with minor gaps       | Selection is limited or partially relevant | Items are unclear or not relevant        |
| Comparison Depth        | 25%       | Provides detailed and comprehensive comparison across multiple aspects        | Provides adequate comparison with some depth    | Limited comparison with few aspects        | Very minimal or no comparison            |
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| <b>Course Code:</b>   | <b>CSA12</b>                                                                                                                                                                                                                                              | <b>Course Title:</b> | <b>Computer Architecture</b>                      |
| <b>CO Assessed:</b>   | <b>CO4</b>                                                                                                                                                                                                                                                | <b>Assessment:</b>   | <b>ASSESSMENT TOOL2-<br/>COMPARATIVE ANALYSIS</b> |
| <b>Weightage:</b>     | <b>30%</b>                                                                                                                                                                                                                                                | <b>Total Marks:</b>  | <b>25</b>                                         |
| <b>CO4 Statement:</b> | <i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i> |                      |                                                   |
| <b>Student Name:</b>  | <b>R DINESH KUMAR</b>                                                                                                                                                                                                                                     | <b>Reg.No.:</b>      | <b>192312258</b>                                  |
| <b>Department:</b>    | <b>B.E ECE</b>                                                                                                                                                                                                                                            | <b>Date:</b>         | <b>24/08/2026</b>                                 |

**ANNEXURE A**  
**COMPARATIVE ANALYSIS**

**1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.**

In a modern processor, the CPU can execute instructions much faster than main memory can supply data. To reduce this difference in speed, processors use a hierarchy of cache memories called **L1, L2 and L3 cache**. Although all three are cache memories, they differ mainly in size, latency, bandwidth and their distance from the processor core.

**L1 cache** is the smallest and fastest cache. It is normally located very close to, or directly associated with, each CPU core. It is generally divided into instruction cache and data cache. Because of its small size, the processor can access it with very low latency. It also provides very high effective bandwidth because the CPU can access it rapidly. The limitation is its capacity. It cannot hold the complete working set of a large application.

**L2 cache** is larger than L1 but slightly slower. It acts as the next level when the required information is not found in L1. Since it has greater capacity, it can store more instructions and data. Its latency is higher than L1, but it is still significantly faster than accessing DRAM. L2 therefore provides a useful balance between capacity and speed.

**L3 cache** is generally much larger than L1 and L2 and is often shared among multiple processor cores. It has higher latency than the first two cache levels, but its larger capacity allows it to retain more frequently used information. It reduces the number of requests that have to go to main memory.

A simple hierarchy can be represented as:

## CPU → L1 → L2 → L3 → DRAM

The important point is that cache performance is not determined by capacity alone. **Cache hit rate** also matters. If the processor finds the required data in L1, it can continue execution with very little delay. If there is an L1 miss but an L2 hit, the additional delay is small compared with going to DRAM. If all cache levels miss, the processor has to wait much longer for main memory.

From a throughput perspective, high cache-hit rates allow the processor to keep its execution units busy. Frequent DRAM accesses can cause stalls and reduce the number of useful instructions completed per unit of time. Therefore, an efficient cache hierarchy improves both processor responsiveness and overall throughput.

However, simply making every cache extremely large is not practical. Larger caches require more chip area, consume more power and may themselves take longer to access. Therefore, L1 should focus on **minimum latency**, L2 on a **balance between speed and capacity**, and L3 on **larger capacity and reducing DRAM traffic**.

In a real processor, these three levels work together rather than competing with one another. The best design is one where frequently accessed data is likely to be found in the faster levels. Therefore, the L1/L2/L3 hierarchy plays an important role in reducing average memory-access time and improving processor performance. This directly matches the assessment's focus on differentiating cache levels based on latency, bandwidth and throughput.

## 2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

SRAM and DRAM are both important types of semiconductor memory, but they are designed for different purposes. The main difference is that **SRAM is faster and more expensive**, while **DRAM provides much greater capacity at a lower cost per bit**. Because of these characteristics, SRAM is mainly used for cache memory, whereas DRAM is commonly used as main memory.

**SRAM, or Static Random Access Memory**, stores each bit using a circuit that does not require periodic refreshing while power is supplied. This allows SRAM to provide very fast access with low latency. It is particularly useful for processor caches, where even a small delay can affect CPU performance.

Another advantage of SRAM is its ability to support high-speed data access and high effective bandwidth. Since cache memory is accessed very frequently, this speed is valuable. However, SRAM requires more transistors per stored bit than DRAM. As a result, it occupies more chip area and is more expensive. This is why processors cannot practically use large amounts of SRAM as their entire main memory.

**DRAM, or Dynamic Random Access Memory**, stores data using a capacitor-based cell and needs periodic refreshing to maintain its contents. Because its cells are smaller, DRAM can provide much higher memory density. This makes it suitable for large main-memory capacities.

Compared with SRAM, DRAM has higher access latency. However, modern DRAM systems can still provide very high bandwidth by using wider interfaces and multiple memory channels.

Therefore, DRAM is not simply “slow memory”; it is high-capacity working memory optimized for large data transfers.

The basic arrangement is:

**CPU → SRAM Cache → DRAM Main Memory**

SRAM allows the processor to quickly access frequently used instructions and data. DRAM stores the larger working set of applications. If the required information is not found in the cache, the processor eventually has to access DRAM, increasing the average access time.

From a throughput point of view, both memories have important roles. SRAM improves processor throughput by reducing waiting time for frequently accessed data. DRAM provides the capacity and bandwidth needed to support large applications and datasets.

There is therefore a clear engineering trade-off. If we use only SRAM, the system would be extremely fast but expensive and limited in capacity. If we use only DRAM, the system would have good capacity but the processor would experience higher memory latency. Combining them provides a much more practical solution.

For example, in a smartphone or laptop, SRAM can be used for L1, L2 and L3 caches, while DRAM can act as the main working memory. This hierarchical approach gives the processor fast access to frequently used information while still providing enough capacity for larger applications.

Therefore, SRAM is best suited to **low-latency cache design**, while DRAM is better suited to **high-capacity main memory**. Their combination is one of the key reasons modern computer systems can achieve both high performance and reasonable cost. The assessment specifically asks for this comparison in terms of latency, bandwidth and throughput.

### **3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.**

Virtual memory and cache memory both improve the usability or performance of a computer system, but they solve different problems. **Cache memory is mainly designed to reduce processor access latency**, whereas virtual memory allows programs to use a large logical address space and manages how memory is mapped between virtual and physical memory.

Cache memory is positioned close to the processor. Frequently accessed instructions and data are copied into cache so that the CPU can retrieve them more quickly. When the required data is found in the cache, the processor avoids accessing slower levels of memory. This is called a **cache hit**.

If the data is not present, a **cache miss** occurs and the system must retrieve it from a lower level of the memory hierarchy. Although this causes additional delay, the penalty is normally much smaller than accessing secondary storage.

Virtual memory works at a different level. It gives each process a virtual address space and maps virtual pages to physical memory. When a required page is not currently present in physical

memory, a **page fault** occurs. The operating system then has to bring that page into memory, potentially from storage.

This means the latency difference can be significant. A cache hit can occur extremely quickly, while a page fault may take much longer because it involves operating-system processing and potentially storage access.

The relationship can be viewed as:

**CPU → Cache → DRAM → Storage**

Cache is used to reduce the average access time between the CPU and memory. Virtual memory and paging manage the relationship between the application's virtual address space and physical memory.

From a throughput perspective, cache memory can significantly improve CPU throughput because it reduces processor stalls. If a program has good **locality of reference**, frequently used data remains in cache and the processor can execute instructions more continuously.

Paging can help system throughput when physical memory is limited because it allows multiple programs to operate without requiring all of their data to remain in RAM simultaneously. However, excessive paging can have the opposite effect. If page faults occur repeatedly, the CPU spends more time waiting for memory pages and the system may experience **thrashing**.

Therefore, cache memory is primarily a **performance optimization**, while virtual memory is primarily a **memory-management and protection mechanism** that can also improve system flexibility.

A practical computer needs both. Cache reduces the processor's average memory-access latency, while virtual memory allows applications to use memory safely and efficiently beyond the limitations of directly addressing physical RAM.

In my view, the most important distinction is that cache tries to make memory access **faster**, while paging tries to make memory **appear larger and more manageable**. Confusing these two mechanisms can lead to an incorrect system design. For high system throughput, the goal should be to maintain a high cache-hit rate while avoiding excessive page faults. This addresses the assessment's required comparison of cache mechanisms and virtual-memory paging in terms of latency and throughput.

#### **4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.**

NAND Flash and DRAM are both widely used memory technologies, but they have very different characteristics. The most important differences are in volatility, latency, bandwidth, capacity and their position in a memory hierarchy.

DRAM is volatile memory, meaning that it loses its stored information when power is removed. It is designed for relatively fast read and write operations and is commonly used as the main memory

of a computer. DRAM provides much lower access latency than NAND Flash and can support high data-transfer bandwidth, especially when multiple memory channels are used.

The main advantage of DRAM is that the processor can directly use it as working memory. Operating systems, applications, databases and active datasets can be placed in DRAM so that the CPU can access them during execution.

NAND Flash, in comparison, is non-volatile. It can retain data even when the system is switched off. This makes it suitable for persistent storage in SSDs, smartphones, memory cards and embedded systems.

The major disadvantage of NAND Flash for active processing is its higher access latency compared with DRAM. It also has different write characteristics and requires management mechanisms such as wear leveling and garbage collection in many storage systems.

The hierarchy can therefore be represented as:

CPU → Cache → DRAM → NAND Flash

DRAM is placed closer to the processor because it is intended for active working data. NAND Flash is placed farther away because it provides persistent, high-capacity storage rather than extremely low-latency processor access.

For example, when a computer starts an application, the application may initially be stored on NAND Flash. The required program and data are loaded into DRAM. Once the data is in DRAM, the processor can access it much more efficiently. Frequently accessed portions may then be placed into cache.

Bandwidth is also important. Modern NAND Flash storage can provide substantial sequential data-transfer rates, especially in SSDs, but its access behavior is different from DRAM. DRAM is better suited to the repeated random-access operations required by active program execution.

From an engineering perspective, it would not make sense to replace DRAM completely with NAND Flash simply because NAND Flash is non-volatile and has high storage density. Doing so would increase memory-access latency and negatively affect processor performance.

At the same time, using DRAM for permanent storage would be impractical because it is volatile and more expensive for large-capacity storage. Therefore, the two technologies complement each other.

A good hierarchical memory system uses DRAM for fast temporary working memory and NAND Flash for large-capacity persistent storage. Cache can then provide an even faster layer above DRAM.

Thus, NAND Flash and DRAM should not be viewed as direct replacements for one another. Their different characteristics make them suitable for different levels of the hierarchy. The assessment specifically requires their comparison based on latency, bandwidth and suitability for hierarchical memory systems.



## 5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

MRAM and RRAM are emerging non-volatile memory technologies that aim to provide alternatives to traditional memory technologies. Both can retain data without continuous power, but they use different physical mechanisms and have different performance characteristics.

**MRAM, or Magnetoresistive Random Access Memory**, stores information using magnetic states rather than relying on electrical charge in the same way as conventional volatile memories. One of its important advantages is non-volatility. Data can remain stored even when power is removed.

MRAM can provide relatively fast access and is attractive for applications that require a combination of speed, persistence and endurance. Because the memory is non-volatile, systems can potentially reduce the need to repeatedly reload certain information after power interruptions.

**RRAM, or Resistive Random Access Memory**, stores information by changing the resistance state of a memory cell. It is also non-volatile and is being investigated for high-density memory and low-power applications.

When comparing access time, both technologies are designed to offer faster access than many conventional non-volatile storage technologies. However, actual performance depends on the particular implementation, memory architecture and manufacturing technology. Therefore, it is not accurate to say that one technology will always be faster in every application.

Throughput depends not only on the individual memory-cell switching speed but also on the architecture surrounding the memory. Factors such as parallelism, memory-controller design, bus width and access pattern can influence overall throughput.

Energy efficiency is another important consideration. Since both MRAM and RRAM are non-volatile, they can potentially reduce energy associated with maintaining stored information. This can be especially useful in battery-powered or edge-computing devices.

A simplified comparison is:

| Feature              | MRAM                               | RRAM                                           |
|----------------------|------------------------------------|------------------------------------------------|
| Data retention       | Non-volatile                       | Non-volatile                                   |
| Access time          | Fast                               | Fast, depending on implementation              |
| Throughput           | Potentially high                   | Potentially high                               |
| Power during standby | Low                                | Low                                            |
| Main advantage       | Speed + endurance + non-volatility | Density + non-volatility + potential low power |



From a practical engineering perspective, the choice should depend on the application. MRAM can be attractive when fast access and endurance are major requirements. RRAM can be attractive when high density, non-volatility and energy efficiency are important.

For example, an embedded device that frequently powers on and off could benefit from non-volatile memory because important information does not have to be reconstructed from slower storage each time. Similarly, edge AI systems could potentially benefit from non-volatile memory to reduce data movement and power consumption.

However, these technologies should not be selected based on one specification alone. Access time, throughput, endurance, energy consumption, density and manufacturing cost should all be considered.

Therefore, both MRAM and RRAM have strong potential in future hierarchical memory systems. The best choice depends on the workload and system requirements rather than simply selecting the technology with the highest individual specification. This matches the assessment requirement to compare MRAM and RRAM using access time, throughput and energy efficiency.

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***Code of Conduct:***

*I R DINESH KUMAR (192312258) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***  
***DINESH KUMAR R***



## MARKS ALLOCATION

|    | Scope of Items<br>(20%) | Comparison<br>Depth (25%) | Use of<br>Evidence<br>(20%) | Critical<br>Thinking (20%) | Clarity of<br>Presentation<br>(15%) |
|----|-------------------------|---------------------------|-----------------------------|----------------------------|-------------------------------------|
| Q1 |                         |                           |                             |                            |                                     |
| Q2 |                         |                           |                             |                            |                                     |
| Q3 |                         |                           |                             |                            |                                     |
| Q4 |                         |                           |                             |                            |                                     |
| Q5 |                         |                           |                             |                            |                                     |

## RUBRICS FOR CALCULATION:

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